

Lumenia — Instawards sprint closeout

Evidence pack · Stellar Türkiye chapter · testnet scope · compiled 2026-08-31

Send USDC by link. The recipient claims it with no wallet, no seed phrase and no gas on their side, because the sponsor covers the fee. Every claim creates and funds a new Stellar account. Lumenia is non-custodial: money is held in escrow on the ledger, never on a Lumenia server.

D1 — Live sponsor service (testnet) COMPLETE

Deployed Cloudflare Worker, reachable now. `/create-account` builds the sponsored sandwich (beginSponsoring → createAccount with startingBalance 0 → changeTrust USDC recipient-sourced → endSponsoring). `/feebump` validates against the anti-drain allowlist, fee-bumps, submits and confirms. Per-IP and per-account rate limiting on both, plus a fee cap enforced before signing.

- Health: lumenia-sponsor.avakit.workers.dev/health
- Sponsored 0-XLM account creation, 4-op sandwich: 43ceea89b034fc6484206348b8ab44fafa4a1349101a63a441cb064a0ace0aa8

D2 — End-to-end walletless claim (testnet) COMPLETE

The sprint's success metric was binary, with no partial credit: one verifiable claim landing USDC in a freshly sponsored 0-XLM account. It is met twice, and both still resolve today.

- Binary-metric claim: b9ef1844c6ca2df732648b965a2f991ba0197643057b2c9e2a60ab52c3e23746 (ledger 3522415)
- Recipient account, still **0.0000000 XLM and 20.0000000 USDC**: GC15ZR6B2TQJDN7VX4TBZAU4J5RBRCKLWYALJEIMPNOM7CTK6AP5PPIR
- Independent second claim on the current stack: 21816364fbe2460ac58c2fcf54dfd24b71f71ad3344f7358dad12d2aa772203
- 60-second demo, recorded on a phone: youtube.com/shorts/X5ie9O7XLYg
- Transaction shown in that video: 029ba09d07ad787697124dac9d8f054fccd203ad16080f1089248932990dc612 — its *Fee Source Account* is the sponsor, not the recipient.

D3 — Anti-drain protection, wired and tested COMPLETE

The validator gates every live `/feebump` unconditionally, before any signature is produced. It checks operation **source and parameters**, not just operation type, and is strict by default: anything the policy does not explicitly permit is rejected. The SOW contracted 14 tests; the suite is now **60/60** and still contains all 14 originals, so it is a superset rather than a substitution.

- Write-up: [ANTI_DRAIN.md](#) · source: `apps/sponsor/src/lib/anti-drain.ts`
- Re-run it yourself: `git clone https://github.com/getlumenia/lumenia && npm install && npm test:antidrain`

Engineering gate

17 offline suites, **539 assertions**, zero failures, no network and no keys required. The escrow contract carries **29 unit and property tests** against a 14-invariant spec, plus fuzzing and mutation testing, and passes clippy under arithmetic-overflow, unwrap and panic lints. All of it runs in CI on every push, together with typechecking, linting and a production build.

Beyond the SOW scope

After this sprint closed, the same flow was proven on Stellar **mainnet with real Circle USDC**, as a hand-approved capped pilot rather than a launch. As of 2026-08-28: 74 approved wallets, 69 accounts opened each with a USDC trustline, 53 funded, **109 transfers with real money**. Caps are \$5 per transfer and \$50 per day, they fail closed if their store is unreachable, and there is a kill switch and a 15-minute watchdog.

- Escrow contract: CAC5JYQ2XEEVJ54EXC7KCG6MTAR05CSUQ2WNKS0M6FALCCU5UTEIWGR4
- First mainnet walletless claim: 1cef173bcd8192b42789ca0dd6bab60f416f18998d24f7b9ad404fd792e6dd0
- Receipt account holding real USDC and 0 XLM: GDXAKZVDWTCZE5ZCESI6IH2NZ6J05TX0VN27F6PLYVQTSSZBISUCXQ55

The cash-out leg was also walked end to end with our own money, putting the total ledger cost of a transfer at roughly **0.145 percent**, against the World Bank's 8.78 percent global average for remittances.

Stated plainly, so nobody has to ask. The escrow contract has had a static-analysis, fuzz and mutation pass but **not a professional audit**, and we do not describe it as audited. The sponsor key is a code-complete AWS KMS integration still running on an environment key in production. The contract owner is a single cold key held offline, not yet a multisig. Those three are the gates we hold ourselves to before raising caps.

Repo github.com/getlumenia/lumenia · Live getlumenia.com · Full evidence file: EVIDENCE.md in the repo, including its own list of deviations from the SOW as written.